1 Maritime Object Detection Datasets and the Annotation Gap

The performance of modern object detection systems is fundamentally constrained by the scale, diversity, and quality of their training data. In the autonomous driving domain, this relationship has been demonstrated repeatedly: each successive generation of benchmark datasets—from KITTI [1] to nuScenes [2] to the Waymo Open Dataset [3]—has enabled corresponding advances in perception algorithms. Maritime autonomy, however, has not benefited from an equivalent data ecosystem. This chapter surveys the landscape of maritime object detection datasets, catalogs their scope and annotation methodology, and quantifies the order-of-magnitude gap that separates maritime data resources from their automotive counterparts. Understanding this gap is essential context for the broader literature review that follows, which examines radar-camera calibration, cross-modal projection, automated dataset construction, sensor fusion, X-band signal processing, and edge deployment—all components of a radar-guided camera labeling pipeline designed to close the maritime annotation deficit.

1.1 The Role of Large-Scale Annotated Datasets in Perception

The deep learning revolution in computer vision was enabled not only by advances in network architectures and computing hardware but also by the availability of large-scale, carefully annotated datasets. ImageNet [4], with its 1.28 million training images spanning 1,000 object categories, demonstrated that supervised pre-training on a sufficiently large corpus could yield feature representations that transferred effectively to downstream tasks. The PASCAL VOC challenge [5] established standardized evaluation protocols that allowed meaningful comparison across detection methods, while MS COCO [6] raised the bar with 330,000 images, 1.5 million object instances, and dense annotations including bounding boxes, segmentation masks, and captions. In the autonomous driving domain, this progression continued: KITTI [1] introduced multi-sensor benchmarks with LiDAR, stereo cameras, and GPS/IMU; nuScenes [2] scaled to 1,000 scenes with full 360-degree sensor coverage and 1.4 million 3D bounding box annotations; and the Waymo Open Dataset [3] reached 12.6 million LiDAR box labels and 11.8 million camera box labels across 1,150 driving scenes.

The impact of dataset scale on detector performance is well documented. Zhang et al. [7] note that large-scale, multi-scenario training data is an indispensable prerequisite for practical deployment of deep learning-based detection systems, and that the maritime domain lacks a widely accepted standardized benchmark of comparable scale. This observation motivates a systematic examination of what maritime datasets do exist, what they contain, and where the critical deficiencies lie.

1.2 A Catalog of Maritime Object Detection Datasets

Maritime detection datasets have emerged from three distinct operational contexts: shore-based surveillance systems monitoring harbors and coastal waters, onboard sensors carried by unmanned surface vehicles (USVs) or autonomous surface vehicles (ASVs), and multi-modal platforms integrating radar, LiDAR, or other active sensors alongside cameras. A fourth category encompasses specialized datasets that employ infrared, hyperspectral, or stereo imaging. The following subsections organize the literature accordingly.

1.2.1 Shore-Based Visual Datasets

Shore-based maritime datasets are typically collected from fixed camera installations overlooking harbors, canals, or coastal waterways. Their primary applications are maritime surveillance, vessel traffic monitoring, and port security.

MarDCT (2015). The Maritime Detection, Classification, and Tracking benchmark [8] was among the earliest publicly available maritime video datasets. Collected by the ARGOS automated surveillance system operating continuously in Venice, Italy, MarDCT comprises 6,742 images and 16 video sequences at 800×240 resolution. Ground truth is provided as bounding boxes for boat detection. While pioneering in its public availability, the dataset's small scale and limited resolution restrict its utility for training modern deep detectors.

Singapore Maritime Dataset (SMD) (2017). Prasad et al. [9] released the Singapore Maritime Dataset alongside a comprehensive survey of video processing for maritime object detection. The dataset contains 81 high-definition (1920×1080) video sequences captured from both onshore (32 sequences) and onboard (4 sequences) viewpoints in Singapore waters, using visual-optical and near-infrared sensors. Annotations cover 10 maritime object classes across diverse environmental conditions including haze, rain, sunrise, midday, and nighttime. SMD established an important early benchmark for evaluating detection and tracking algorithms in tropical maritime environments.

SeaShips (2018). Shao et al. [10] constructed a large-scale, precisely annotated dataset from approximately 10,080 real-world video segments captured by a deployed coastline surveillance system. SeaShips contains 31,455 images covering six common vessel types: ore carrier, bulk cargo carrier, general cargo ship, container ship, fishing boat, and passenger ship. Images were carefully selected to represent variations in scale, hull visibility, illumination, viewpoint, background complexity, and occlusion. Each image is annotated with ship type labels and high-precision bounding boxes. At the time of its release, SeaShips was among the largest publicly available maritime detection datasets and remains widely used as a benchmark.

MCShips (2020). Zheng and Zhang [11] introduced MCShips, a dataset of 14,709 annotated images collected via web crawlers and search engines, spanning 13 fine-grained ship categories (6 warship types and 7 civilian vessel types). Unlike surveillance-derived datasets, MCShips captures a wide diversity of imaging conditions, perspectives, and resolutions found in publicly available online imagery. Each image is annotated with bounding boxes and fine-grained class labels, supporting both detection and classification tasks.

ABOShips (2021). Iancu et al. [12] constructed the ABOShips dataset from video sequences recorded in the Finnish Archipelago using a camera mounted on a waterbus. The dataset comprises 9,880 images containing 41,967 annotated object instances across 9 vessel categories plus seamarks and miscellaneous floaters. A two-stage annotation process was employed: initial manual labeling followed by automated tracking (CSRT tracker) to identify and correct labeling inconsistencies. Benchmark evaluations with Faster R-CNN, R-FCN, SSD, and EfficientDet established baseline performance levels, with Faster R-CNN (Inception-ResNet v2 backbone) achieving the highest overall accuracy.

MCMOD (2023). Sun et al. [13] released the Multi-Category Maritime Object Detection dataset, consisting of 16,166 images with 98,590 annotated objects across 10 maritime categories including fishing vessels, speedboats, engineering ships, cargo ships, yachts, sailboats, buoys, rafts, cruise ships, and an "others" class. Three onshore cameras operating continuously in Hainan, China, captured high-resolution (1920×1080) video under varied conditions including fog, rain, and evening illumination. The cameras' rotation and zoom capabilities produced diverse scenes, and the dataset's emphasis on small-scale distant targets addresses a persistent challenge in maritime detection.

KOLOMVERSE (2024). Nanda et al. [14] introduced the Korea Open Large-Scale Image Dataset for Object Detection in the Maritime Universe, representing the largest publicly available maritime detection dataset to date. Drawn from 5,845 hours of video data captured across 21 territorial waters of South Korea, KOLOMVERSE contains 186,419 4K-resolution (3840×2160) images with 732,039 annotated object instances across five classes: ship (393,936), buoy (34,080), fishnet buoy (95,815), lighthouse (60,362), and wind farm (147,846). The dataset is partitioned into training (149,175), validation (18,643), and test (18,601) splits. KOLOMVERSE's scale and resolution represent a significant step forward for maritime detection, though its five-class taxonomy and shore-based perspective limit its applicability to fine-grained classification and onboard perception tasks.

1.2.2 USV/ASV-Mounted Datasets

Datasets collected from the perspective of an unmanned or autonomous surface vehicle present unique challenges not captured by shore-based imagery: low camera elevation above the waterline, dynamic horizon orientation due to wave-induced platform motion, frequent spray and lens contamination, and the need to detect obstacles at close range for collision avoidance.

MaSTr1325 (2019). Bovcon et al. [15] released a maritime semantic segmentation training dataset containing 1,325 diverse images captured over two years by a USV operating in the Gulf of Koper, Slovenia. Each image is annotated at pixel level with three semantic classes: sea, sky, and environment (encompassing all obstacles and shoreline). While the three-class taxonomy is coarse, the dataset was instrumental in training early deep segmentation models for USV obstacle detection. The annotation procedure required approximately twenty minutes per image, underscoring the labor intensity of pixel-level maritime labeling.

FloW (2021). Cheng et al. [16] presented FloW, the first dataset targeting floating waste detection in inland waterways, accepted at ICCV 2021. FloW comprises two sub-datasets: FloW-Img, an image collection with bounding box annotations for floating debris, and FloW-RI, a multimodal sub-dataset containing synchronized millimeter-wave radar data and camera images. The inclusion of radar data makes FloW one of the earliest maritime-adjacent datasets to provide cross-modal supervision signals. Baseline experiments demonstrated that floating waste detection remains challenging due to small target size, strong water-surface reflections, and bankside clutter.

MassMIND (2023). Nirgudkar et al. [17] contributed the Massachusetts Maritime Infrared Dataset, containing 2,916 long-wave infrared (LWIR) images captured over two years in and around Boston Harbor. The dataset is segmented using both instance and semantic segmentation into seven classes, making it the first maritime dataset to provide fine-grained LWIR segmentation labels for both living (humans, marine wildlife) and non-living (vessels, structures) obstacles. Evaluation with UNet, PSPNet, and DeepLabv3 architectures established baseline segmentation performance for thermal maritime imagery.

LaRS (2023). Žust et al. [18] released the first maritime panoptic obstacle detection benchmark, accepted at ICCV 2023. LaRS contains over 4,000 per-pixel labeled keyframes with nine preceding temporal context frames each, totaling more than 40,000 frames. Annotations span 8 “thing” classes, 3 “stuff” classes, and 19 global scene attributes. The dataset features the largest diversity among maritime datasets in recording locations (lakes, rivers, and seas), scene types, obstacle classes, and acquisition conditions. An online evaluation server with results from 27 segmentation methods provides a standardized benchmark. LaRS represents the current state of the art for USV-perspective semantic and panoptic segmentation evaluation.

WaterScenes (2024). Zhang et al. [19] introduced WaterScenes, the first multi-task 4D radar-camera fusion dataset for autonomous driving on water surfaces. Collected by a USV equipped with a 4D millimeter-wave radar and a monocular camera, the dataset comprises 54,120 synchronized image-radar frame pairs covering more than 200,000 annotated objects. Annotations support five tasks: object detection, instance segmentation, semantic segmentation, free-space segmentation, and waterline segmentation. Environmental diversity spans daytime, nightfall, and nighttime conditions; normal, dim, and strong lighting; sunny, overcast, rainy, and snowy weather; and four waterway types (river, lake, canal, moat). WaterScenes is notable for being the first maritime dataset to provide both pixel-level camera annotations and point-level radar annotations, enabling direct evaluation of radar-camera fusion algorithms in the maritime domain.

SeePerSea (2024). Jeong et al. [20] released the first publicly accessible multi-modal perception dataset for in-water obstacle detection from ASVs, featuring synchronized LiDAR point clouds and RGB images. Collected over four years across locations in the United States, Barbados, and South Korea, SeePerSea contains 11,561 annotated frames with navigation-oriented three-class labels (ship, buoy, other). The dataset is significant as the first maritime LiDAR-camera dataset with object labels across both modalities, supporting evaluation of cross-modal fusion for aquatic obstacle detection.

1.2.3 Multi-Modal Radar/LiDAR/Camera Datasets

The most recent generation of maritime datasets integrates multiple active and passive sensing modalities, reflecting the recognition that robust maritime perception requires sensor fusion. These datasets are also the most directly relevant to radar-guided camera labeling, as they provide synchronized radar and camera data that can serve as ground truth for evaluating cross-modal projection and automatic annotation pipelines.

Pohang Canal Dataset (2023). Chung et al. [21] released a comprehensive multi-modal maritime dataset collected along a 7.5 km route through the Pohang Canal, inner and outer ports, and near-coastal areas in South Korea. The sensor suite includes stereo cameras, an infrared camera, an omnidirectional camera, three LiDARs, a marine radar, GPS, and an attitude heading reference system. Data were collected under diverse weather and visibility conditions. The Pohang Canal Dataset is among the most sensor-rich maritime collections available, though it is primarily designed for navigation and SLAM rather than object detection, and its annotation density for detection tasks is limited.

M^{2}SODAI (2023). Jang et al. [22] presented a multi-modal maritime object detection dataset combining RGB and hyperspectral image sensors, accepted at NeurIPS 2023 Datasets and Benchmarks track. The dataset provides synchronized aerial image pairs with bounding box annotations across multiple maritime object categories. Hyperspectral imaging (HSI), which captures over 100 spectral channels in the visible and near-infrared range, enables extraction of material-specific signatures that can discriminate maritime objects from sea-surface clutter even when visual appearance is ambiguous. M^{2}SODAI is the first maritime dataset to incorporate hyperspectral sensing, addressing the false-positive challenge inherent in single-modality visual detection over dynamic water surfaces.

PoLaRIS (2024). Choi et al. [23] released the PoLaRIS (Pohang Canal Object Detection and Tracking Dataset in Maritime Environments) dataset, accepted at ICRA 2025. PoLaRIS provides approximately 360,000 labeled images from RGB and thermal infrared cameras, along with point-wise annotations from radar and LiDAR. It is the first dataset offering multi-modal annotations specifically tailored to maritime object detection and tracking, including tracking ground truth for dynamic objects with bounding boxes projected across modalities. The dataset includes annotations for objects as small as 10×10 pixels, reflecting the prevalence of small, distant targets in maritime environments. With approximately 190,000 detection-grade bounding box annotations, PoLaRIS represents the most densely annotated multi-sensor maritime dataset available.

MOANA (2024). Jang et al. [24] introduced the Multi-Radar Dataset for Maritime Odometry and Autonomous Navigation Application, integrating short-range LiDAR, medium-range W-band radar, and long-range X-band radar into a unified multi-scale sensing framework. The dataset comprises 7 sequences collected from diverse coastal regions with varying navigation difficulty. MOANA is the first maritime dataset to combine multiple radar frequency bands, enabling research on multi-scale maritime perception from close-range obstacle detection to long-range situational awareness. Its primary focus is odometry estimation, SLAM, and place recognition rather than object detection, and the limited number of sequences constrains its statistical power for training data-hungry detection models.

1.2.4 Multi-Spectral and Specialized Datasets

Several datasets address maritime perception through non-standard imaging modalities or specialized acquisition configurations.

VAIS (2015). Zhang et al. [25] released the Visible and Infrared Ship dataset, the first publicly available collection of paired visible and thermal infrared maritime imagery. VAIS contains over 1,000 paired RGB-IR images across six vessel categories (merchant, sailing, passenger, medium-other, tug, and small), all acquired from pier-side vantage points. The simultaneous dual-band acquisition enables research on cross-spectral recognition and day-night vessel classification, though the dataset's modest scale limits its applicability for training deep networks.

MARVEL (2016). Gundogdu et al. [26] constructed a large-scale maritime vessel image dataset by harvesting approximately 2 million user-uploaded images and metadata from an online ship-spotting community. Images are categorized into 26 superclasses constructed from 109 fine-grained vessel type labels, with additional attributes including vessel identity, photograph category, and year of construction. MARVEL supports vessel classification, verification, retrieval, and recognition tasks. However, the dataset's web-sourced provenance means images are captured under uncontrolled conditions without standardized annotations, and no bounding box or segmentation labels are provided, limiting its utility for detection tasks.

MODS (2022). Bovcon et al. [27] released a USV-oriented object detection and obstacle segmentation benchmark containing approximately 81,000 stereo images synchronized with onboard IMU measurements, with over 60,000 annotated objects. MODS addresses two major perception tasks—maritime object detection and maritime obstacle segmentation—and introduces a standardized evaluation protocol that measures detection accuracy in terms meaningful for practical USV navigation. Benchmark results from 19 state-of-the-art detection and segmentation methods provide a comprehensive performance landscape for USV perception.

1.3 Comprehensive Dataset Comparison

Table 1 presents a systematic comparison of the maritime datasets surveyed in this chapter. The table reveals several patterns: (1) dataset scale has grown substantially over the past decade, from thousands to hundreds of thousands of images; (2) the majority of datasets remain single-modality visual collections; (3) multi-modal datasets with radar or LiDAR are a recent development (2023–2024) and remain small by comparison; and (4) annotation granularity varies widely, from image-level class labels to pixel-level panoptic segmentation.

Several observations emerge from this comparison. First, KOLOMVERSE [14] is the only maritime dataset to exceed 100,000 detection-annotated images, yet it covers only five object classes from a shore-based perspective. Second, WaterScenes [19] is the only dataset providing both dense camera annotations and radar point annotations from a vessel-mounted platform, making it uniquely relevant for radar-camera fusion research. Third, the multi-sensor datasets most suitable for evaluating radar-guided labeling—Pohang [21], PoLaRIS [23], and MOANA [24]—either lack dense detection annotations or comprise only a handful of sequences.

1.4 The Automotive Benchmark: Scale and Infrastructure

To contextualize the maritime annotation gap, it is instructive to examine the scale of automotive perception benchmarks that have driven progress in self-driving vehicle perception. Table 2 summarizes the key automotive datasets.

The automotive ecosystem benefits from a virtuous cycle: large datasets enable better models, which attract more investment, which funds further data collection and annotation. The Waymo Open Dataset alone contains 12.6 million LiDAR 3D bounding box labels with tracking identifiers across 1,150 scenes, each spanning 20 seconds of driving with well-synchronized and calibrated LiDAR and camera data [3]. nuScenes provides 1.4 million 3D bounding box annotations across 1,000 scenes with full 360-degree sensor coverage including six cameras, five radars, and one LiDAR [2]. The automotive radar-camera datasets RADIATE [28] and the Oxford Radar RobotCar Dataset [29] further demonstrate the automotive community's investment in multi-modal data collection.

1.5 Quantifying the Maritime–Automotive Gap

Table 3 presents a direct quantitative comparison between the best-resourced maritime and automotive datasets across several key metrics. The gaps are stark and consistent across all dimensions.

The most critical gap is in multi-sensor data volume. The Waymo Open Dataset provides 1,150 fully annotated multi-sensor scenes, and nuScenes adds another 1,000, for a combined total exceeding 2,000 scenes with synchronized LiDAR, camera, and radar data.

Table 1: Comprehensive comparison of maritime object detection datasets. Annotation types: BBox = bounding box, Seg = semantic segmentation, Inst = instance segmentation, Pan = panoptic segmentation, PW = point-wise, Trk = tracking ID.

Dataset	Year	Sensors	Images	Objects	Classes	Annotation
MarDCT [8]	2015	RGB	6,742	—	Boat	BBox
VAIS [25]	2015	RGB + IR	2,000	1,242	6	BBox
MARVEL [26]	2016	RGB (web)	2,000,000	—	26	Class label
SMD [9]	2017	RGB + NIR	81 vid	—	10	BBox (video)
SeaShips [10]	2018	RGB	31,455	40,077	6	BBox
MaSTr1325 [15]	2019	RGB + IMU	1,325	—	3	Seg (pixel)
MCShips [11]	2020	RGB (web)	14,709	14,709	13	BBox
ABOShips [12]	2021	RGB	9,880	41,967	9+	BBox
FloW [16]	2021	RGB + mmW radar	2,000+	6,000+	Waste	BBox + PW
MODS [27]	2022	Stereo + IMU	81,000	60,000+	Multi	BBox + Seg
MassMIND [17]	2023	LWIR	2,916	—	7	Seg + Inst
LaRS [18]	2023	RGB	40,000+	—	8+3	Pan (pixel)
M^{2}SODAI [22]	2023	RGB + HSI	5,764+	—	Multi	BBox
Pohang [21]	2023	Multi (7+)	—	—	—	Nav. labels
MCMOD [13]	2023	RGB	16,166	98,590	10	BBox
KOLOMVERSE [14]	2024	RGB	186,419	732,039	5	BBox
WaterScenes [19]	2024	RGB + 4D radar	54,120	200,000+	Multi	BBox + Seg + PW
PoLaRIS [23]	2024	RGB+IR+LiDAR+R	360,000	∼190k	Multi	BBox + PW + Trk
SeePerSea [20]	2024	RGB + LiDAR	11,561	—	3	BBox + PW
MOANA [24]	2024	Multi-radar+L	7 seq.	—	—	Nav. labels

Table 2: Major automotive perception benchmark datasets for comparison with maritime equivalents.

Dataset	Year	Sensors	Images	Labels	Classes	Annotation
ImageNet [4]	2012	RGB	1,281,167	—	1,000	Class label
KITTI [1]	2012	Stereo+LiDAR	14,999	200,000+	8	3D BBox
MS COCO [6]	2014	RGB	330,000	1,500,000	80	BBox + Seg
nuScenes [2]	2020	6C+5R+1L	1,400,000	1,400,000	23	3D BBox + Trk
Waymo [3]	2020	5C+5L	—	12,600,000	4	3D BBox + Trk

Table 3: Order-of-magnitude gap between automotive and maritime dataset resources across key metrics. Gap factor is computed as the ratio of automotive to maritime values.

Metric	Automotive Best	Maritime Best	Gap Factor
Detection images (single dataset)	1,281,167 (ImageNet)	186,419 (KOLOMVERSE)	∼7×
3D bbox labels (single dataset)	12,600,000 (Waymo)	∼190,000 (PoLaRIS)	∼66×
Multi-sensor scenes	2,150 (Waymo+nuScenes)	7 (MOANA)	∼307×
Annotation instances (total)	>15,000,000	∼1,300,000	∼12×
Sensor modalities per dataset (max)	12 (nuScenes)	7+ (Pohang/PoLaRIS)	∼2×
Detection classes (max, single)	80 (COCO)	13 (MCShips)	∼6×

By comparison, MOANA [24]—the most comprehensive multi-radar maritime dataset—comprises only 7 sequences. Even aggregating all maritime multi-modal datasets (Pohang, PoLaRIS, MOANA, WaterScenes, FloW, SeePerSea), the total number of independently collected multi-sensor sequences remains well below 100. This approximately 300-fold gap in multi-sensor scene diversity represents perhaps the most consequential deficiency for developing robust maritime perception systems, as sensor fusion algorithms require diverse training examples to generalize across the wide range of environmental conditions encountered at sea.

The 3D bounding box label gap is similarly severe. Waymo's 12.6 million LiDAR-derived 3D labels dwarf the approximately 190,000 detection-grade annotations in PoLaRIS [23], yielding a gap factor of roughly 66. This disparity is compounded by the fact that maritime 3D labels are almost exclusively derived from 2D bounding boxes (sometimes augmented with range information), whereas automotive 3D labels benefit from dense LiDAR point clouds that enable precise 3D cuboid fitting.

Aggregating across all maritime datasets surveyed in this chapter, the total number of detection-grade annotated maritime images (those with bounding box or segmentation annotations) is approximately 700,000. This figure is dominated by KOLOMVERSE at 186,419, followed by PoLaRIS at approximately 360,000 (though many of these are repetitive keyframes from overlapping sequences), MODS at 81,000, WaterScenes at 54,120, SeaShips at 31,455, and the remaining datasets collectively contributing under 100,000. For comparison, MS COCO alone provides 330,000 annotated images with 1.5 million object instances [6], and the Waymo Open Dataset encompasses over 200,000 annotated LiDAR frames [3].

1.6 Annotation Cost and Quality Challenges

The maritime annotation gap reflects not merely a shortage of collection infrastructure but also the disproportionate difficulty and cost of annotating maritime imagery compared to terrestrial scenes.

1.6.1 Domain Expertise Requirements

Maritime object annotation demands specialized knowledge that general-purpose crowd workers typically lack. Distinguishing between vessel types—cargo ships, tankers, fishing trawlers, naval vessels, recreational craft—requires familiarity with hull forms, superstructure configurations, and operational context that is not readily acquired without maritime training. Zhang et al. [7] emphasize that domain-specific expertise is a prerequisite for constructing high-quality maritime training data, noting that misclassification at the labeling stage propagates directly into detector training as systematic label noise. The ABOShips annotation pipeline [12], which required a second pass using visual tracking to identify and correct initial labeling errors, illustrates the additional quality-control overhead inherent in maritime annotation.

1.6.2 Small and Distant Targets

Maritime scenes routinely contain targets at ranges of several kilometers, appearing as clusters of only a few pixels in the image. MCMOD [13] explicitly addresses the prevalence of small-scale objects, and PoLaRIS [23] includes annotations for objects as small as 10×10 pixels. Annotating such targets is time-consuming and error-prone: annotators must zoom to high magnification to delineate object boundaries, and the low signal-to-noise ratio for distant targets increases inter-annotator disagreement. The annotation time per image for maritime datasets is correspondingly high; Bovcon et al. [15] report approximately 20 minutes per image for pixel-level semantic segmentation of MaSTr1325, compared to typical rates of 5–10 minutes for road-scene segmentation in automotive datasets.

1.6.3 Ambiguous Boundaries and Visual Clutter

Water-surface conditions create annotation ambiguities that are largely absent from terrestrial scenes. Vessel wakes, bow waves, and spray obscure hull boundaries; sun glare and specular reflections from the water surface produce bright artifacts that can be confused with objects; and atmospheric effects (haze, fog, rain) reduce contrast between objects and background. These factors increase the variance of bounding box placement across annotators. Additionally, partially submerged objects, overlapping vessels at marinas, and the lack of fixed spatial reference points (unlike lane markings or curbs in driving scenes) further complicate boundary delineation.

1.6.4 Class Imbalance and Long-Tail Distribution

Maritime traffic exhibits severe class imbalance. In KOLOMVERSE [14], ships account for 393,936 of 732,039 instances (53.8%), while buoys contribute only 34,080 (4.7%). Similar distributions appear in other datasets: small recreational craft dominate in coastal waters, while large commercial vessels are relatively rare. This long-tail distribution means that even large datasets may contain too few examples of safety-critical minority classes (e.g., kayaks, swimmers, semi-submerged debris) to train reliable detectors.

1.7 Domain-Specific Perception Challenges

Beyond annotation difficulty, the maritime domain presents perception challenges that necessitate larger and more diverse training sets than might be required for a comparably complex terrestrial task.

1.7.1 Environmental Variability

Maritime environments exhibit extreme variability across temporal, meteorological, and geographic dimensions. Sea state ranges from mirror-calm to storm conditions with multi-meter waves; lighting spans from direct tropical sun (with intense specular reflections) to overcast polar twilight; visibility ranges from unlimited to near-zero in fog, rain, or snow. WaterScenes [19] explicitly captures variation across daytime, nightfall, and nighttime conditions; sunny, overcast, rainy, and snowy weather; and four distinct waterway types. KOLOMVERSE [14] samples from 21 territorial waters to achieve geographic diversity. Despite these efforts, no single maritime dataset approaches the environmental coverage that automotive datasets achieve through crowd-sourced collection across entire nations and seasons.

1.7.2 Horizon and Background Complexity

The maritime horizon, while seemingly simple, creates distinctive detection challenges. Objects transitioning from below to above the horizon change dramatically in background context (dark water to bright sky or vice versa). Coastal backgrounds introduce shoreline structures, breakwaters, and moored vessels that create complex clutter patterns. In narrow waterways, such as those captured in the Pohang Canal Dataset [21], the visual scene more closely resembles an urban canyon than open ocean, requiring detectors to handle both maritime and semi-terrestrial visual contexts.

1.7.3 Scale and Aspect Ratio Diversity

The range of object scales in maritime scenes exceeds that in typical driving scenarios. A single image may contain both a large vessel filling much of the frame and distant targets comprising fewer than 100 pixels. Moreover, the aspect ratios of maritime objects vary more widely than those of vehicles, pedestrians, and cyclists: a container ship viewed broadside may have an aspect ratio exceeding 8:1, while a small buoy is approximately 1:1. This scale diversity challenges both anchor-based and anchor-free detection architectures, which must maintain sensitivity across several orders of magnitude in object area.

1.7.4 Temporal Dynamics and Motion Patterns

Wave-induced platform motion on USV-mounted cameras produces rapid and largely unpredictable changes in camera orientation, leading to motion blur, horizon oscillation, and intermittent spray occlusion. The MODS dataset [27] and LaRS [18] both capture these dynamics through temporally consecutive frame sequences. Additionally, the relative motion between an observing platform and maritime targets differs qualitatively from automotive scenarios: closing rates can be very slow (vessels approaching head-on at a few knots) or very fast (high-speed craft), and lateral apparent motion is dominated by the observer's heading changes rather than lane-based geometry.

1.8 Summary and Research Gaps

This chapter has surveyed 20 maritime object detection datasets spanning a decade of development (2015–2024), organized by collection platform and sensor modality. Several structural factors account for the persistent annotation gap between maritime and automotive domains:

1. **Limited collection infrastructure.** Automotive datasets benefit from fleets of sensor-equipped vehicles operating on public roads; maritime data collection requires access to vessels, coastal installations, or USVs that are less widely available and more expensive to operate.

2. **Fewer research groups.** The autonomous driving community encompasses dozens of major industrial laboratories (Waymo, Cruise, Argo AI, Mobileye, Baidu Apollo) and hundreds of academic groups, each contributing data and benchmarks. The maritime perception community is substantially smaller, with fewer organizations possessing both the sensing infrastructure and the annotation resources to produce large-scale datasets.

3. **Environmental diversity.** Maritime environments span open ocean, coastal waters, rivers, lakes, canals, and harbors, each with distinct visual characteristics, traffic patterns, and object taxonomies. Achieving representative coverage requires data

collection across many geographically and climatically distinct locations, a logistical challenge that compounds the per-image annotation cost.

4. **Annotation difficulty.** As documented in Section 1.6, the combination of small distant targets, ambiguous boundaries, domain expertise requirements, and severe class imbalance makes maritime annotation two to four times slower per image than automotive annotation. At typical annotation service rates, the cost of manually labeling a maritime dataset to Waymo-equivalent scale (12 million bounding boxes) would be prohibitive for most research groups.

Against this backdrop, several emerging approaches offer pathways to narrow the gap:

1. **Radar-guided camera labeling.** By projecting radar detections into the camera frame, bounding box annotations can be generated automatically without human intervention. The availability of WaterScenes [19], PoLaRIS [23], and MOANA [24] as multi-modal reference datasets provides initial validation targets for such pipelines. The noise characteristics of radar-projected labels are systematic (arising from calibration and projection geometry) rather than random, which suggests that domain-specific denoising strategies may be required.

2. **AIS-assisted annotation.** The Automatic Identification System (AIS) provides vessel identity, position, heading, and speed for cooperative maritime traffic. Fusing AIS tracks with camera imagery enables automatic association of detected objects with vessel metadata, facilitating both bounding box generation (via projected AIS positions) and fine-grained class labeling (via vessel type codes transmitted in AIS messages).

3. **Active learning and selective annotation.** Rather than annotating all collected frames uniformly, active learning strategies can identify the most informative frames for human annotation, maximizing detection improvement per annotation dollar. The temporal redundancy inherent in maritime video (vessels move slowly relative to frame rate) suggests that aggressive frame selection could substantially reduce annotation volume without sacrificing model performance.

4. **Cross-modal label transfer.** Beyond radar-to-camera projection, cross-modal transfer encompasses LiDAR-to-camera (as explored in SeePerSea [20]), thermal-to-visible (as enabled by VAIS [25] and PoLaRIS [23]), and hyperspectral-to-RGB (as proposed in M^{2}SODAI [22]). Each modality pair offers distinct advantages: radar provides range and velocity; LiDAR provides precise 3D geometry; thermal provides illumination-invariant detection; and hyperspectral provides material discrimination. A comprehensive automated labeling framework may ultimately combine multiple cross-modal signals to produce annotations that exceed the quality achievable by any single modality alone.

The remainder of this literature review examines each of these technical components in detail, beginning with radar-camera sensor registration and calibration (Section **??**), progressing through radar-to-camera projection and bounding box generation (Section **??**), and extending to automated dataset construction methodologies (Section **??**), radar-camera fusion (Section **??**), X-band signal processing (Section **??**), and edge computing for real-time deployment (Section **??**).

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